PHYS4055 Week 2 Tutorial Solutions

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Question 1: Beam-splitter transformation

Consider two independent bosonic modes with annihilation operators $\hat{a}$ and $\hat{b}$, satisfying

$$[\hat{a}, \hat{a}^\dagger] = [\hat{b}, \hat{b}^\dagger] = 1, \qquad [\hat{a}, \hat{b}] = [\hat{a}, \hat{b}^\dagger] = 0.$$

A lossless beam splitter is described by the unitary operator

$$\hat{U}_B = \exp\left[\theta\left(\hat{a}^\dagger\hat{b} - \hat{a}\hat{b}^\dagger\right)\right].$$

The Baker–Campbell–Hausdorff relation may be written as

$$e^{\hat{A}}\hat{B}e^{-\hat{A}} = \hat{B} + [\hat{A}, \hat{B}] + \frac{1}{2!}[\hat{A}, [\hat{A}, \hat{B}]] + \frac{1}{3!}[\hat{A}, [\hat{A}, [\hat{A}, \hat{B}]]] + \cdots .$$

Using this relation:

1. Calculate the first four nested commutators required to evaluate

$$\hat{U}_B\hat{a}\hat{U}_B^\dagger \quad \text{and} \quad \hat{U}_B\hat{b}\hat{U}_B^\dagger.$$

2. Group the resulting series into sine and cosine expansions and show that

$$\hat{U}_B\hat{a}\hat{U}_B^\dagger = \hat{a}\cos\theta - \hat{b}\sin\theta,$$

$$\hat{U}_B\hat{b}\hat{U}_B^\dagger = \hat{a}\sin\theta + \hat{b}\cos\theta.$$

3. Defining the beam-splitter transmissivity as

$$\eta = \cos^2\theta,$$

show that the output-mode operators are

$$\hat{a}' = \sqrt{\eta}\,\hat{a} - \sqrt{1-\eta}\,\hat{b},$$

$$\hat{b}' = \sqrt{1-\eta}\,\hat{a} + \sqrt{\eta}\,\hat{b}.$$

Solution

Define

$$\hat{K} = \hat{a}^\dagger\hat{b} - \hat{a}\hat{b}^\dagger,$$

so that

$$\hat{U}_B = e^{\theta\hat{K}}, \qquad \hat{U}_B^\dagger = e^{-\theta\hat{K}}.$$

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Using

$$e^{\theta\hat{K}}\hat{A}e^{-\theta\hat{K}} = \hat{A} + \theta[\hat{K},\hat{A}] + \frac{\theta^2}{2!}[\hat{K},[\hat{K},\hat{A}]] + \frac{\theta^3}{3!}[\hat{K},[\hat{K},[\hat{K},\hat{A}]]] + \cdots,$$

we first consider the transformation of $\hat{a}$.

The first commutator is

$$\begin{aligned}[\hat{K},\hat{a}] &= [\hat{a}^\dagger\hat{b} - \hat{a}\hat{b}^\dagger, \hat{a}] \\ &= [\hat{a}^\dagger\hat{b},\hat{a}] - [\hat{a}\hat{b}^\dagger,\hat{a}] \\ &= -\hat{b}.\end{aligned}$$

We also require

$$\begin{aligned}[\hat{K},\hat{b}] &= [\hat{a}^\dagger\hat{b} - \hat{a}\hat{b}^\dagger, \hat{b}] \\ &= \hat{a}.\end{aligned}$$

The nested commutators for $\hat{a}$ are therefore

$$\begin{aligned}[\hat{K},\hat{a}] &= -\hat{b}, \\ [\hat{K},[\hat{K},\hat{a}]] &= -\hat{a}, \\ [\hat{K},[\hat{K},[\hat{K},\hat{a}]]] &= \hat{b}, \\ [\hat{K},[\hat{K},[\hat{K},[\hat{K},\hat{a}]]]] &= \hat{a}.\end{aligned}$$

Substituting these into the BCH expansion gives

$$\begin{aligned}\hat{U}_B\hat{a}\hat{U}_B^\dagger &= \hat{a} - \theta\hat{b} - \frac{\theta^2}{2!}\hat{a} + \frac{\theta^3}{3!}\hat{b} + \frac{\theta^4}{4!}\hat{a} - \cdots \\ &= \hat{a}\left(1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \cdots\right) - \hat{b}\left(\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \cdots\right) \\ &= \hat{a}\cos\theta - \hat{b}\sin\theta.\end{aligned}$$

For the transformation of $\hat{b}$, the nested commutators are

$$\begin{aligned}[\hat{K},\hat{b}] &= \hat{a}, \\ [\hat{K},[\hat{K},\hat{b}]] &= -\hat{b}, \\ [\hat{K},[\hat{K},[\hat{K},\hat{b}]]] &= -\hat{a}, \\ [\hat{K},[\hat{K},[\hat{K},[\hat{K},\hat{b}]]]] &= \hat{b}.\end{aligned}$$

The BCH expansion then gives

$$\begin{aligned}\hat{U}_B\hat{b}\hat{U}_B^\dagger &= \hat{b} + \theta\hat{a} - \frac{\theta^2}{2!}\hat{b} - \frac{\theta^3}{3!}\hat{a} + \frac{\theta^4}{4!}\hat{b} + \cdots \\ &= \hat{b}\left(1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \cdots\right) + \hat{a}\left(\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \cdots\right) \\ &= \hat{a}\sin\theta + \hat{b}\cos\theta.\end{aligned}$$

Therefore,

$$\hat{a}' = \hat{a}\cos\theta - \hat{b}\sin\theta, \qquad \hat{b}' = \hat{a}\sin\theta + \hat{b}\cos\theta.$$

Defining the beam-splitter transmissivity by

$$\eta = \cos^2\theta,$$

we have

$$\cos\theta = \sqrt{\eta}, \qquad \sin\theta = \sqrt{1-\eta}.$$

Thus,

$$\boxed{\hat{a}' = \sqrt{\eta}\,\hat{a} - \sqrt{1-\eta}\,\hat{b}}$$

and

$$\boxed{\hat{b}' = \sqrt{1-\eta}\,\hat{a} + \sqrt{\eta}\,\hat{b}.}$$

A. Question 2: Energy conservation at a beam splitter

Consider a lossless beam splitter acting on two optical modes. The input and output mode operators are related by

$$\hat{a}' = \sqrt{\eta}\,\hat{a} - \sqrt{1-\eta}\,\hat{b},$$

and

$$\hat{b}' = \sqrt{1-\eta}\,\hat{a} + \sqrt{\eta}\,\hat{b},$$

where $0 \le \eta \le 1$ is the beam-splitter transmissivity.

The photon-number operators of the two output modes are

$$\hat{N}_{a'} = \hat{a}'^{\dagger}\hat{a}', \qquad \hat{N}_{b'} = \hat{b}'^{\dagger}\hat{b}'.$$

1. Write down the transformed creation operators $\hat{a}'^{\dagger}$ and $\hat{b}'^{\dagger}$.
2. Calculate $\hat{N}_{a'}$ explicitly in terms of the input-mode operators $\hat{a}$, $\hat{a}^{\dagger}$, $\hat{b}$, and $\hat{b}^{\dagger}$.
3. Similarly, calculate $\hat{N}_{b'}$ in terms of the input-mode operators.
4. By adding your results, show that

   $$\hat{N}_{a'} + \hat{N}_{b'} = \hat{N}_a + \hat{N}_b,$$

   where

   $$\hat{N}_a = \hat{a}^{\dagger}\hat{a}, \qquad \hat{N}_b = \hat{b}^{\dagger}\hat{b}.$$

Solution

1. Taking the Hermitian conjugate of the output-mode operators gives

   $$\hat{a}'^{\dagger} = \sqrt{\eta}\,\hat{a}^{\dagger} - \sqrt{1-\eta}\,\hat{b}^{\dagger},$$

   and

   $$\hat{b}'^{\dagger} = \sqrt{1-\eta}\,\hat{a}^{\dagger} + \sqrt{\eta}\,\hat{b}^{\dagger}.$$

2. The photon-number operator for output mode a' is

   $$\hat{N}_{a'} = \hat{a}'^{\dagger}\hat{a}'.$$

   Substituting the beam-splitter transformations,

   $$\begin{aligned}\hat{N}_{a'} &= \left(\sqrt{\eta}\,\hat{a}^{\dagger} - \sqrt{1-\eta}\,\hat{b}^{\dagger}\right)\left(\sqrt{\eta}\,\hat{a} - \sqrt{1-\eta}\,\hat{b}\right) \\ &= \eta\,\hat{a}^{\dagger}\hat{a} - \sqrt{\eta(1-\eta)}\,\hat{a}^{\dagger}\hat{b} - \sqrt{\eta(1-\eta)}\,\hat{b}^{\dagger}\hat{a} + (1-\eta)\,\hat{b}^{\dagger}\hat{b}.\end{aligned}$$

   Therefore,

   $$\boxed{\hat{N}_{a'} = \eta\,\hat{N}_a + (1-\eta)\,\hat{N}_b - \sqrt{\eta(1-\eta)}\left(\hat{a}^{\dagger}\hat{b} + \hat{b}^{\dagger}\hat{a}\right).}$$

3. The photon-number operator for output mode b' is

$$\hat{N}_{b'} = \hat{b}'^{\dagger}\hat{b}'.$$

Substituting the beam-splitter transformations,

$$\begin{aligned}\hat{N}_{b'} &= \left(\sqrt{1-\eta}\,\hat{a}^{\dagger} + \sqrt{\eta}\,\hat{b}^{\dagger}\right)\left(\sqrt{1-\eta}\,\hat{a} + \sqrt{\eta}\,\hat{b}\right) \\ &= (1-\eta)\,\hat{a}^{\dagger}\hat{a} + \sqrt{\eta(1-\eta)}\,\hat{a}^{\dagger}\hat{b} + \sqrt{\eta(1-\eta)}\,\hat{b}^{\dagger}\hat{a} + \eta\,\hat{b}^{\dagger}\hat{b}.\end{aligned}$$

Therefore,

$$\boxed{\hat{N}_{b'} = (1-\eta)\,\hat{N}_a + \eta\,\hat{N}_b + \sqrt{\eta(1-\eta)}\left(\hat{a}^{\dagger}\hat{b} + \hat{b}^{\dagger}\hat{a}\right)}.$$

4. Adding the two output photon-number operators gives

$$\begin{aligned}\hat{N}_{a'} + \hat{N}_{b'} &= \eta\,\hat{N}_a + (1-\eta)\,\hat{N}_b - \sqrt{\eta(1-\eta)}\left(\hat{a}^{\dagger}\hat{b} + \hat{b}^{\dagger}\hat{a}\right) \\ &\quad + (1-\eta)\,\hat{N}_a + \eta\,\hat{N}_b + \sqrt{\eta(1-\eta)}\left(\hat{a}^{\dagger}\hat{b} + \hat{b}^{\dagger}\hat{a}\right) \\ &= [\eta + (1-\eta)]\,\hat{N}_a + [(1-\eta) + \eta]\,\hat{N}_b \\ &= \hat{N}_a + \hat{N}_b.\end{aligned}$$

Hence,

$$\boxed{\hat{N}_{a'} + \hat{N}_{b'} = \hat{N}_a + \hat{N}_b}.$$

The beam splitter can redistribute photons between the two modes, but it preserves the total photon number.

B. Question 3: Partial trace

A single photon passes through a balanced beam splitter, producing the two-mode state

$$|\psi\rangle_{ab} = \frac{1}{\sqrt{2}}\left(|1\rangle_a|0\rangle_b + |0\rangle_a|1\rangle_b\right).$$

1. Write down the two-mode density operator

$$\hat{\rho}_{ab} = |\psi\rangle_{ab}\langle\psi|.$$

2. Suppose that we only have access to mode a. Calculate its reduced density operator by tracing over mode b:

$$\hat{\rho}_a = \mathrm{Tr}_b(\hat{\rho}_{ab}) = \sum_{n=0}^{1} {}_b\langle n|\hat{\rho}_{ab}|n\rangle_b.$$

3. What is the probability of detecting zero photons in mode a? What is the probability of detecting one photon?

4. Is the reduced state $\hat{\rho}_a$ pure or mixed?

Solution

1. The two-mode density operator is

$$\hat{\rho}_{ab} = |\psi\rangle_{ab}\langle\psi|.$$

Substituting the two-mode state gives

$$\begin{aligned}\hat{\rho}_{ab} &= \frac{1}{2}\left(|1\rangle_a|0\rangle_b + |0\rangle_a|1\rangle_b\right)\left({}_a\langle 1|{}_b\langle 0| + {}_a\langle 0|{}_b\langle 1|\right) \\ &= \frac{1}{2}\Big(|1\rangle_a|0\rangle_{ba}\langle 1|_b\langle 0| + |1\rangle_a|0\rangle_{ba}\langle 0|_b\langle 1| \\ &\quad + |0\rangle_a|1\rangle_{ba}\langle 1|_b\langle 0| + |0\rangle_a|1\rangle_{ba}\langle 0|_b\langle 1|\Big).\end{aligned}$$

Using the shorthand

$$|n,m\rangle = |n\rangle_a|m\rangle_b,$$

this can also be written as

$$\hat{\rho}_{ab} = \frac{1}{2}\left(|1,0\rangle\langle 1,0| + |1,0\rangle\langle 0,1| + |0,1\rangle\langle 1,0| + |0,1\rangle\langle 0,1|\right).$$

2. The reduced density operator of mode a is

$$\hat{\rho}_a = \mathrm{Tr}_b(\hat{\rho}_{ab}).$$

In the basis $\{|0\rangle_b, |1\rangle_b\}$,

$$\hat{\rho}_a = {}_b\langle 0|\hat{\rho}_{ab}|0\rangle_b + {}_b\langle 1|\hat{\rho}_{ab}|1\rangle_b.$$

The first term is

$${}_b\langle 0|\hat{\rho}_{ab}|0\rangle_b = \frac{1}{2}|1\rangle_{aa}\langle 1|.$$

The second term is

$${}_b\langle 1|\hat{\rho}_{ab}|1\rangle_b = \frac{1}{2}|0\rangle_{aa}\langle 0|.$$

The cross terms vanish because

$$\langle 0|1\rangle = 0.$$

Therefore,

$$\boxed{\hat{\rho}_a = \frac{1}{2}|0\rangle_{aa}\langle 0| + \frac{1}{2}|1\rangle_{aa}\langle 1|}.$$

3. The probability of detecting zero photons in mode a is

$$\begin{aligned}P(0) &= {}_a\langle 0|\hat{\rho}_a|0\rangle_a \\ &= \frac{1}{2}.\end{aligned}$$

The probability of detecting one photon is

$$\begin{aligned}P(1) &= {}_a\langle 1|\hat{\rho}_a|1\rangle_a \\ &= \frac{1}{2}.\end{aligned}$$

4. To determine whether the state is pure or mixed, we calculate its purity:

$$\begin{aligned}\mathrm{Tr}(\hat{\rho}_a^2) &= \mathrm{Tr}\left[\frac{1}{4}|0\rangle_{aa}\langle 0| + \frac{1}{4}|1\rangle_{aa}\langle 1|\right] \\ &= \frac{1}{2}.\end{aligned}$$

Since

$$\mathrm{Tr}(\hat{\rho}_a^2) < 1,$$

the reduced state is mixed. Thus,

$$\boxed{\hat{\rho}_a \text{ is a mixed state.}}$$